

AMARNATH PAARA

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Summary

Senior Software Engineer with 4+ years of experience designing and delivering scalable, high-performance software solutions across insurance, ad-tech, and fintech domains. Strong expertise in backend development using Java, Spring Boot, JAX-RS, Node.js, Express, C, and C++, with proven experience building RESTful and event-driven microservices using Apache Kafka. Proficient in frontend engineering with React.js, AngularJS, JavaScript (ES6+), and jQuery, delivering responsive, user-centric web applications. Hands-on experience with Docker, Kubernetes (EKS), Jenkins CI/CD, and AWS (EC2, S3, RDS) to enable cloud-native deployments and automated delivery pipelines. Adept at system optimization, database performance tuning, SQL development, security implementation (RBAC, KYC integrations), and modernizing legacy systems, consistently driving measurable improvements in scalability, reliability, and engineering productivity.

Skills

- **Programming Languages:** Java 8/11/17, Python, C++17, JDBC, Servlets, JSP, JavaScript (ES6+), TypeScript, SQL, PL/SQL, HTML5, CSS3
- **Enterprise Frameworks & Spring Ecosystem:** Spring Boot, Spring MVC, Spring Cloud, Spring Core (IoC, AOP), Spring Data JPA, Spring Security, Spring Batch, Hibernate ORM, JPA, Flask, FastAPI
- **Cloud & Infrastructure Engineering:** AWS, Docker, Kubernetes (EKS, Orchestration, Helm), Jenkins, CI/CD Pipeline Architecture, Containerization, Microservices Deployment, Auto-scaling, NGINX
- **Backend & Systems Architecture:** Microservices Architecture, RESTful API Design & Development, SOAP Web Services, Event-Driven Architecture, Apache Kafka, WebSockets, API Gateways, gRPC, Distributed Systems
- **Methodologies & Application Servers:** Agile & Scrum, CI/CD, WebSphere Application Server, Java Mail Sessions, Test-Driven Development (TDD)
- **Frontend & Full-Stack Engineering:** Angular, React.js, Single Page Applications (SPA), Component-Based Architecture, Responsive Web Design, jQuery, JSP, Qt Framework Data, Cache & Performance Engineering: Redis, Distributed Caching, Oracle, Database Design, Query Optimization & Tuning, PL/SQL Stored Procedures, HQL, Hibernate Criteria, Data Integrity, Connection Pooling, Pandas, NumPy, Scikit learn, SQLite, ODBC
- **Software Engineering & QA:** Unit & Integration Testing (JUnit, Mockito, Mocha, Cucumber, pytest, Google Test), Code Quality (SonarQube), Logging & Monitoring (Kibana), Code Refactoring, Performance Tuning, JVM Tuning, Memory Management, Multithreading
- **Libraries & SDKs:** STL, Boost, C++ REST SDK (Casablanca), Apache Spark, TensorFlow, PyTorch
- **Core Competencies:** Full-Stack Development, System Architecture & Design, Legacy Modernization, Performance Optimization, Cloud-Native Development, Real-Time Data Processing, Machine Learning Pipelines, High-Frequency Systems, Technical Leadership, Cross-Functional Collaboration, Agile Delivery, Mentoring & Code Reviews.

Experience

Senior Software Engineer

02/2025 – Present

PNC

USA

- Architected Spring Boot-based microservices to decompose legacy insurance policy management systems, to enable independent service scaling, reducing inter-service latency by 30% for claims and underwriting platforms.
- Engineered high-throughput REST APIs using Java 17 and Spring MVC, to support real-time policy lifecycle operations, processing 1M+ monthly transactions for enterprise insurance clients.
- Designed event-driven workflows with Apache Kafka, to asynchronously process claims, notifications, and audit events, improving system resiliency and eliminating synchronous bottlenecks for downstream services.
- Implemented containerized deployments using Docker and Kubernetes (EKS), to standardize runtime environments, accelerating release cycles by 35% for cross-functional development teams.
- Optimized cloud infrastructure on AWS (EC2, RDS, S3), to improve fault tolerance and auto-scaling behavior, reducing infrastructure costs by 25% for production workloads.
- Developed secure backend integrations using Node.js and Express, to interface with third-party compliance and identity providers, strengthening data protection for regulated insurance workflows.
- Constructed CI/CD pipelines with Jenkins and Git-based workflows, to automate build, test, and deployment stages, cutting production deployment failures by 40% for engineering operations.
- Enhanced application observability using centralized logging and metrics, to proactively detect system anomalies, reducing mean time to resolution (MTTR) for customer-impacting incidents.

Software Engineer

08/2021 – 08/2023

InMobi

India

- Developed scalable backend services using Java, Spring Boot, and RESTful APIs, to power ad-tech data ingestion pipelines, supporting billions of daily ad events for global advertisers.
- Implemented real-time data processing with Apache Kafka, to stream campaign performance metrics, enabling near real-time analytics dashboards for marketing and sales stakeholders.
- Built interactive front-end components using React.js and JavaScript (ES6+), to visualize campaign insights, improving customer engagement and dashboard responsiveness by 20%.
- Refactored performance-critical modules written in C++, to optimize memory usage and execution paths, reducing service crashes and latency under peak traffic conditions.

- Containerized microservices using Docker, to standardize development environments, decreasing onboarding time for new engineers from days to hours.
- Automated deployment workflows through Jenkins CI pipelines, to ensure consistent releases across staging and production, shortening release cycles by 30%.
- Optimized SQL and NoSQL data access patterns, to improve query throughput and reduce response times for reporting services used by internal analytics teams.

Software Developer

08/2020 – 07/2021

Insight Global

India

- Engineered RESTful web services using Java, JAX-RS, and Spring Boot, to support client-facing financial applications, handling 1M+ API calls per month for end-users.
- Constructed dynamic UI components using JavaScript, jQuery, and AngularJS, to translate complex business workflows into user-friendly web interfaces for financial clients.
- Refactored legacy C and C++ modules, to eliminate memory leaks and undefined behavior, improving application stability and reducing production incidents by 90%.
- Integrated third-party KYC and identity verification APIs, to automate customer onboarding processes, saving 5+ hours daily for operations and compliance teams.
- Developed optimized SQL queries and stored procedures, to generate financial and compliance reports, enabling faster decision-making for business analysts and leadership.
- Implemented role-based access control (RBAC), to safeguard sensitive financial data, ensuring adherence to internal security and compliance standards.
- Introduced Jenkins-based CI pipelines, to automate builds and testing, improving code reliability and reducing manual deployment errors for small engineering teams.

Educations

Master of Science: Information Technology

2024

University of Cincinnati